

Guide: Building Your First RAG Application with Amazon Bedrock

I'm going to walk you through how to set up a **Retrieval-Augmented Generation (RAG)** system from scratch. This setup allows an AI model to use your specific, private documents to answer questions accurately.

Follow these steps to build your own knowledge-driven application.

Phase 1: Securing Your Environment

First, I want you to set up a dedicated environment. We should avoid using your root account for daily tasks to keep your AWS environment secure.

1. **Create an IAM User:** Navigate to the **IAM Console**. Create a new user with a name (e.g., `project-developer-2026`).
 2. **Enable Console Access:** Select "Provide user access to the AWS Management Console." I recommend setting a custom password and unchecking "Users must create a new password at next sign-in" to save time during this build.
 3. **Set Permissions:** Attach the `AdministratorAccess` policy directly to this user. This ensures you won't run into permission errors while configuring Bedrock and OpenSearch.
 4. **Log In Safely:** Copy the sign-in URL provided at the end of the process. Open a **Private/Incognito window** and log in as this new user.
-

Phase 2: Preparing Your Knowledge Source

Now, we need a place to store the data that the AI will learn from.

1. **Check Your Region:** Ensure you are in a region where Bedrock is available, such as **North Virginia (us-east-1)**.
 2. **Create an S3 Bucket:** Go to the S3 console and create a bucket with a unique name (e.g., `my-project-repository-data`).
 3. **Upload Your File:** Upload the document you want the AI to reference. This can be a PDF, TXT, or MD file containing the information you want the model to "know."
-

Phase 3: Building the Knowledge Base

This is where we connect the AI to your data.

1. **Initiate Knowledge Base:** In the **Amazon Bedrock** console, go to **Knowledge bases** and click **Create**. Select "Knowledge base with vector store."
 2. **Assign a Name:** Give it an name like **KB-Core-Logic**.
 3. **Service Role:** Choose the option to let AWS create a new service role automatically. This handles the behind-the-scenes "handshake" between Bedrock and S3.
 4. **Configure the Data Source:** * Select **Amazon S3** as your data source.
 - Name the source (e.g., **Primary-DS**).
 - Browse S3 and select the specific file you uploaded in Phase 2.
-

Phase 4: Setting Processing Strategies

I'll show you how to tell Bedrock how to "read" and "organize" your data.

Parsing Strategy

I recommend starting with the **Default Parser**. It's the most cost-efficient way to extract text from your files. If your documents have very complex tables, you can explore other options later, but for now, keep it simple.

Chunking Strategy

The AI can't read a whole book at once; it needs "chunks."

- **My Recommendation:** Use **Default Chunking**.
 - **Why?** It automatically splits your text into pieces of roughly 300 tokens. This is usually the best balance for ensuring the model gets enough context without getting overwhelmed.
-

Phase 5: Creating the Vector Database

For the AI to search your data, we must turn text into numbers (vectors).

1. **Select Embeddings Model:** Choose an embeddings model like **Amazon Titan Embeddings G1**.
 2. **Vector Store Setup:** Choose the "Quick create a new vector store" option. This will automatically provision an **Amazon OpenSearch Serverless** collection for you.
 3. **Retention Policy:** Under advanced settings, I suggest keeping the data deletion policy on **Retain** so your work isn't lost if you change settings later.
-

Phase 6: Syncing and Testing

Your infrastructure is ready, but the data isn't in the database yet.

1. **Sync the Data:** Once the knowledge base is created, go to the data source section and click **Sync**. This "teaches" the system your data by converting the S3 file into vectors.
 2. **Run a Test:** Click **Test knowledge base**.
 3. **Select a Model:** Choose a model for the response (e.g., **Amazon Nova Micro**).
 4. **Ask Questions:** Type questions into the chat. If the setup is correct, the AI will provide answers based specifically on the document you uploaded.
-

Phase 7: Cleaning Up (Crucial Step)

To avoid being charged for resources you aren't using, I want you to follow this exact deletion order:

- **Step 1:** Delete the **Knowledge Base** in Bedrock. Make sure to check the box to delete the vector data.
- **Step 2:** Go to **OpenSearch Service**. Delete the **Serverless Collection**, then manually delete the **Access, Encryption, and Network policies**.
- **Step 3:** Go to **S3**. Empty your bucket and then delete the bucket itself.
- **Step 4:** Log back into your root account and delete the **IAM User** you created.